Supplementary Material

for the paper

"*Counting HyperGraphlets via Color Coding: a Quadratic Barrier and How to Break It*"

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Contents

1 Proofs

This section contains the proofs of results stated in the main paper that were omitted due to space constraints.

1.1 Lemma 5

Lemma 5. *NW-IE solves NW in time* $O(2^{\Delta(H)} \cdot |V|)$.

Proof. Fix $v \in V$ and $X \subseteq E(v)$. Note that line 4 increments $t[X]$ by $w(v)$ if and only if $v \in \cap_{e \in C} e$. Therefore, at line 5 we have:

$$t[X] = \sum_{v \in \bigcap X} w(v) = w\Big(\bigcap X\Big). \tag{1}$$

By construction, then, the second loop ensures that for any $v \in V$:

$$\begin{aligned}
\eta(v) &= \sum_{\emptyset \neq X \subseteq E(v)} (-1)^{|X|+1} t[X] - w(v) && (2)\\
&= \sum_{\emptyset \neq X \subseteq E(v)} (-1)^{|X|+1} w\left(\bigcap X\right) - w(v) && \text{by Eq. (1)} \quad (3)\\
&= w\left(\bigcup E(v)\right) - w(v) && \text{by I.-E. (Eq. (14))} \quad (4)
\end{aligned}$$

For the time complexity, for every $v \in V$ the algorithm performs at most $2^{|E(v)|} \leq 2^{\Delta(H)}$ operations, each of which takes time $O(1)$. □

1.2 Lemma 6

Lemma 6. *If* H *is* $(\alpha, \beta)-$*nice, Algorithm 3 computes the counters of Equation (4) in time* $2^{O(k)} \cdot \big(2^{\beta}|V| + \alpha^2\beta\|H\|\big)$.

Proof. The correctness follows from the discussion above, which shows that at line 19 we have $\eta(v) = \mathcal{C}(T_2, S_2, v)$, from $\mathcal{C}(T, S, v) = \frac{1}{d}\sum_{\substack{S_1, S_2 \subset S \\ S_1 \cap S_2 = \emptyset}} \mathcal{C}(T_1, S_1, v) \sum_{u \sim v} \mathcal{C}(T_2, S_2, u)$, and by construction of the algorithm. Let us bound the running time.

By Lemma 4, line 5 takes time $O(\|H\|)$. By the discussion above, line 6 and line 7 take time $O(\alpha^2\beta\|H\|)$. The same discussion showed that each execution of the loop of line 14 takes time take time $O(\alpha^2\beta\|H\|)$ as well, while line 11, line 12 and line 13 take time respectively $O(|V|)$, $O(\alpha^2|E|)$, and $O(2^{\beta}|V|)$. Thus, every iteration of the loop of line 10 takes time

$$O\left(2^{\beta}|V| + \alpha^2|E| + \alpha^2\beta\|H\|\right) = O\left(2^{\beta}|V| + \alpha^2\beta\|H\|\right). \tag{5}$$

which already dominates the running time of the rest. Noting that the said loop makes $2^{O(k)}$ iterations completes the proof. □

1.3 Lemma 7

Lemma 7. *Given as input* (T, S, v), SAMPLENEIGH *returns* (T_2, S_2, u) *such that* $u \in N(v)$ *with probability proportional to* $C(T_2, S_2, u)$.

Proof. Consider any iteration of the loop of line 4. Let $p(u)$ be the probability, ignoring line 12, that the iteration draws the particular vertex u. We shall prove that:

$$p(u) \propto \mathcal{C}(S_2, u) \cdot (\mathbb{1}\{u \in N_{\leq\alpha}(v)\} + \mathbb{1}\{u \in N_{>\alpha}(v)\}) \tag{6}$$

By line 12, this implies that every single iteration draws *and* accepts u with probability proportional to $C(T_2, S_2, u)$, proving the claim.

Let $p_{\leq\alpha}(u)$ be the probability that an execution of line 6 draws u, and $p_{>\alpha}(u)$ the probability that an execution of the block of line 7 draws u. By the law of total probability:

$$p(u) = p(N_{\leq\alpha}(v)) \cdot p_{\leq\alpha}(u) + p(N_{>\alpha}(v)) \cdot p_{>\alpha}(u) \tag{7}$$

$$= \frac{W_{\leq\alpha}(v) \cdot p_{\leq\alpha}(u)}{W_{\leq\alpha}(v) + W_{>\alpha}(v)} + \frac{W_{>\alpha}(v) \cdot p_{>\alpha}(u)}{W_{\leq\alpha}(v) + W_{>\alpha}(v)} \tag{8}$$

Let us then analyse $p_{\leq\alpha}(u)$ and $p_{>\alpha}(u)$ separately.

For $p_{\leq\alpha}(u)$, line 6 implies:

$$p_{\leq\alpha}(u) = \frac{\mathcal{C}(S_2, u)}{W_{\leq\alpha}(v)} \cdot \mathbb{1}\{u \in N_{\leq\alpha}(v)\} \tag{9}$$

For $p_{>\alpha}(u)$, let $p_{inn}(u)$ be the probability that, ignoring line 11, one iteration of line 8 draws a particular vertex u; let $p(e)$ be the probability that one execution of line 9 returns the specific hyperedge e; and let $p(u|e)$ be the probability that, conditional on this event, one execution of line 10 returns u. By the law of total probability:

$$p_{inn}(u) = \mathbb{1}\{u \in N_{>\alpha}(v)\} \cdot \sum_{e \in E(v)} p(u \mid e)\, p(e) \tag{10}$$

$$= \mathbb{1}\{u \in N_{>\alpha}(v)\} \tag{11}$$

$$\cdot \sum_{e \in E(v)} \frac{\mathcal{C}(S_2, u)}{\mathcal{C}_2(S_2, e)} \frac{\mathcal{C}(S_2, e)}{\sum_{e' \in E(v)} \mathcal{C}(S_2, e')} \cdot \mathbb{1}\{u \in e\} \tag{12}$$

$$= \mathbb{1}\{u \in N_{>\alpha}(v)\} \cdot \sum_{e \in E(v)} \frac{\mathcal{C}(S_2, u)}{\sum_{e' \in E(v)} \mathcal{C}(S_2, e')} \cdot \mathbb{1}\{u \in e\} \tag{13}$$

$$= \mathbb{1}\{u \in N_{>\alpha}(v)\} \cdot \frac{\mathcal{C}(S_2, u)|E(v) \cap E(u)|}{\sum_{e' \in E(v)} \mathcal{C}(S_2, e')} \tag{14}$$

By line 11, this implies that every single iteration draws and accepts u with probability proportional to $\mathcal{C}(S_2, u) \cdot \mathbb{1}\{u \in N_{>\alpha}(v)\}$, that is:

$$p_{>\alpha}(u) \propto \mathcal{C}(S_2, u) \cdot \mathbb{1}\{u \in N_{>\alpha}(v)\} \tag{15}$$

Since $\sum_{u \in N(v)} \mathcal{C}(S_2, u) \cdot \mathbb{1}\{u \in N_{>\alpha}(v)\} = W_{>\alpha}(v)$, we deduce that:

$$p_{>\alpha}(u) = \frac{\mathcal{C}(S_2, u)}{W_{>\alpha}(v)} \cdot \mathbb{1}\{u \in N_{>\alpha}(v)\} \tag{16}$$

Plugging $p_{\leq\alpha}(u)$ and $p_{>\alpha}(u)$ in Equation (8) proves the claim. □

1.4 Lemma 8

Lemma 8. *With the random number generators precomputed as above, each invocation of* SampleNeigh *takes expected time* $O(\beta^2)$.

Proof. Let I be the random variable that counts the total number of iterations executed by the while loop at line 8. It holds $I \sim Geom(|E(v) \cap E(u)|)$. Let T be the random variable that counts the total number of iterations within the body of a single execution of the while loop at line 4. Then:

$$T = \begin{cases} 1 & \text{with probability } p(N_{\leq\alpha}) \\ I & \text{with probability } 1 - p(N_{\leq\alpha}) \end{cases}$$

It follows that

$$\mathbb{E}[T] = p(N_{\leq\alpha}) + (1 - p(N_{\leq\alpha}))\mathbb{E}[I] \leq 1 + \beta \tag{17}$$

Thus, it holds $\mathbb{E}[T] = O(\beta)$. Let K be the random variable that counts total number of times the loop at line 4 is executed, and N the random variable that counts the total number of executions (including all the loops). Consider now another version of Algorithm 4 where line 12 accepts with fixed probability $\frac{1}{2}$, and let K', N' be the equivalent of K, N for this version. By a simple coupling argument, $K \leq K'$ and $N \leq N'$, and clearly K' is independent of the variables T_i. By Wald's equality, then,

$$N \leq N' = \sum_{i=1}^{K'} T_i = \mathbb{E}[K']\mathbb{E}[T] \leq 2\mathbb{E}[T] = O(\beta)\,. \tag{18}$$

Now let us bound the time taken by the execution of every line. Outside the loop at line 4, the execution of each line takes at most $O(1)$. Drawing u at line 6 takes $O(1)$. The same holds for line 10. Sampling a hyperedge at line 9 takes time $O(1)$, whereas computing $|E(v) \cap E(u)|$ at line 11 takes time at most $O(\beta)$. Thus, if the expected number of iterations is $O(\beta)$, it follows that each invocation of SampleNeigh takes expected time $O(\beta^2)$. □

1.5 Lemma 9

Lemma 9. *By spending an additional $O(\alpha^k \cdot |E_{\leq\alpha}|)$ preprocessing time, one can compute the k-hypergraphlet $H[U]$ induced by any given k-vertex $U \subseteq V$ in time $2^{O(k)} \cdot \beta$.*

Proof. Given U, we check for each $X \subseteq U$ with $|X| \geq 2$ whether there exists $e \in E$ such that $e \cap U = X$, in which case we add X to the hyperedges of $H[U]$. We now discuss how e can be found in a generic set of edges E; we then show how to apply the idea to $E_{\leq\alpha}$ and $E_{>\alpha}$. Suppose we have precomputed, for every $X \subseteq U$, the number $N[X] = |\{e \in E : X \subseteq e\}|$. Moreover let $N^*(X, U) = |\{e \in E : e \cap U = X\}|$; note that $N^*(X, U) > 0$ means $e \cap U = X$ for some $e \in E$. Now, by inclusion-exclusion,

$$N^*(X, U) = \sum_{Y \subseteq U \setminus X} (-1)^{|Y|} N[X \cup Y]\,. \tag{19}$$

Therefore, given the counters $N[X]$, we can compute $N^*(X, U)$ in time $2^{O(|U|)} \leq 2^k$.

Let us now discuss how to implement this for $E_{\leq\alpha}$ and $E_{>\alpha}$. For $E_{\leq\alpha}$, in the build-up phase for every $e \in E_{\leq\alpha}$ we enumerate all subsets $X \subseteq e$ of at most k vertices, increasing $N[X]$ accordingly. This computes the counters $N[X]$ explicitly in time $O(\alpha^k \cdot |E_{\leq\alpha}|)$. For $E_{>\alpha}$, observe that $N[X] = \left|\bigcup_{v \in X} E_{>\alpha}(v)\right|$, and the right-hand side can be computed in time $O(k\beta)$, yielding a total time of $O(2^k \cdot k\beta)$ for computing all counters once given U. By a final bound, given U, checking whether e exists in $E = E_{\leq\alpha} \cup E_{>\alpha}$ takes a total time of $2^{O(k)} \cdot \beta$, as claimed. □

1.6 Lemma 10

Lemma 10. *Given a subset $U \subseteq V$ of k vertices, one can compute the adjacency matrix A_U of $\mathsf{Gaif}(H[U])$ in time $O(k^2(\beta + \ln |V|))$.*

Proof. Clearly, $A_U[u][v] = 1$ if and only if $u \in N_{\leq\alpha}(v)$ or $E_{>\alpha}(u) \cap E_{>\alpha}(v) \neq \emptyset$. The condition $u \in N_{\leq\alpha}(v)$ can be verified in time $O(\log |V|)$ via binary search as $N_{\leq\alpha}(v)$ is sorted and has size at most $|V|$. The condition $E_{>\alpha}(u) \cap E_{>\alpha}(v) \neq \emptyset$ can be verified in time $O(\beta)$ since both sets are sorted and have size at most β. A union bound over all k^2 pairs yields the claim. □

1.7 Corollary 1

Corollary 1. *Algorithm 5 takes, on each invocation, expected time $O(k\beta^2)$. Moreover, the vertex set of the output treelet is $U \in \mathcal{V}_k(H)$ with probability proportional to the number of spanning trees $\sigma(H[U])$ of* $\mathsf{Gaif}(H[U])$.

Proof. The running time bound follows from Lemma 8 and the fact that SAMPLE makes $k-1$ calls to SAMPLENEIGH, together with the bounds on the time for drawing (T, v). The claim on the distribution follows from Lemma 7 and the same analysis MOTIVO[1] □

1.8 Theorem 1

This section of the appendix is dedicated to the proof of Theorem 1. In particular, we prove the following equivalent statement.

Theorem 1. *There exists an FPT-reduction from k-CLIQUE to k-SH that, for every instance (G, k) of k-CLIQUE, yields an instance (H, k') of k-SH with the following properties:*

- $k' = \binom{k}{2}(k+1) \leq \frac{k^3}{2}$.
- *H is $(k^2, 0)$-nice.*

Moreover, the reduction runs in time $O\left(k^2 \cdot (|E(G)| + |V(G)|)\right)$.

This clearly implies that k-SH is $W[1]$-hard. The original claim follows from the well-known lower bound $n^{\Omega(k)}$ of k-CLIQUE under ETH [2]. Note that, by the second item, k-SH remains $W[1]$-hard and is subject to a $n^{\Omega(\sqrt[3]{\kappa})}$ ETH-conditional lower bound even if one takes into account the (α, β)-niceness of the hypergraph by adopting as a parameter $\kappa = k + \alpha + \beta$.

The rest of this section gives the proof of Theorem 1.

The reduction. Let (G, k) be an instance of k-CLIQUE. We construct the corresponding instance (H, k') of k-SH as follows. For each vertex $v \in V(G)$, create a set $Q_v = \{a_v^1, \ldots, a_v^{\binom{k}{2}}\}$, and for each edge $e \in E(G)$, create a singleton $R_e = \{c_e\}$. Define the hypergraph H by

$$V(H) = \bigcup_{v \in V(G)} Q_v \ \cup \bigcup_{e \in E(G)} R_e \tag{20}$$

and

$$E(H) = \{\, E_{uv} : e = \{u, v\} \in E(G),\ E_{uv} = Q_u \cup Q_v \cup R_e \,\}. \tag{21}$$

Finally, we set $k' := (k+1)\binom{k}{2}$, and we take (H, k') as the output instance of k-SH.

See Figure 1 for a visual representation of the reduction. Intuitively, one can also think of this construction as encoding a weighted graph G' on the same vertex and edge set as G, where each vertex has weight $\binom{k}{2}$ and each edge weight 1.

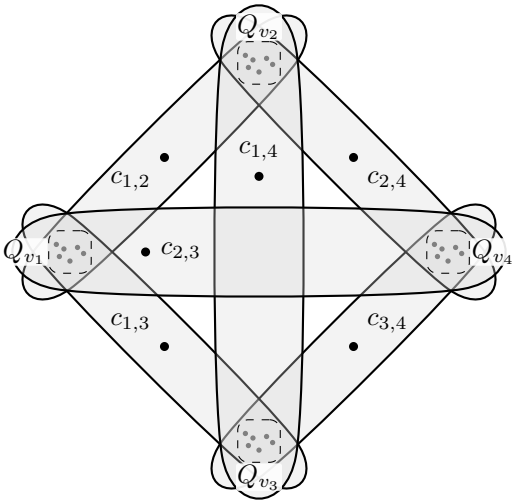


Figure 1: The hypergraph H obtained by the reduction of Theorem 1 on $G = K_4$.

We now prove that the construction above is indeed an FPT reduction that satisfies the claim of Theorem 1.

Claim 2. *The construction above runs in time $O\left(k^2 \cdot (|E(G)| + |V(G)|)\right)$.*

Proof. Let (G, k) be an instance of k-Clique. For each vertex $v \in V(G)$ we create the set Q_v of size $\binom{k}{2}$, which takes a total of $O(k^2 \cdot |V(G)|)$ time. Then, for each edge $e = \{u, v\} \in E(G)$ we create the vertex c_e and the hyperedge $E_{uv} = Q_u \cup Q_v \cup \{c_e\}$, of size $2\binom{k}{2} + 1 = O(k^2)$, for a total time of $O(k^2 \cdot |E(G)|)$. Summing these two bounds gives the claimed running time $O(k^2 \cdot (|V(G)| + |E(G)|))$. □

The next observation is useful to prove the correctness of the reduction. From now on, $H[U]$ refers to the section hypergraph induced by $U \subseteq V(H)$, as formalized in Definition 2.3.

Observation 1. *Let (G, k) be an instance of k-Clique and (H, k') the instance of k-SH obtained via the reduction of Theorem 1. Let $U \subseteq V(H)$ be such that $H[U]$ is connected and $|U| \geq 2$. Then:*

- *For every $v \in V(G)$, either $Q_v \cap U = \emptyset$ or $Q_v \subseteq U$.*
- *For every $e = \{u, v\} \in E(G)$, if $c_e \in U$ then $Q_u \cup Q_v \subseteq U$.*

Proof. For the first item, suppose for a contradiction that there exists $v \in V(G)$ such that $Q_v \cap U \neq \emptyset$ but $Q_v \nsubseteq U$. Take any $x \in Q_v \cap U$ and any $y \in Q_v \setminus U$. Every hyperedge of H containing x is of the form $E_{vw} = Q_v \cup Q_w \cup R_e$, where $e = \{v, w\}$ for some neighbor w of v in G. However, since $y \notin U$, no such hyperedge E_{vw} is entirely contained in U, and therefore no hyperedge containing x appears in $H[U]$. Hence x is an isolated vertex in $H[U]$. As $|U| \geq 2$ and $H[U]$ is connected, this is impossible. Thus either $Q_v \subseteq U$ or $Q_v \cap U = \emptyset$.

For the second item, if $c_e \in U$ but some vertex of Q_u or Q_v is not in U, then the hyperedge $E_{u,v} = Q_u \cup Q_v \cup R_e$ is not entirely contained in U. Thus no hyperedge of $H[U]$ contains c_e, and c_e would be isolated in $H[U]$. Again this contradicts the connectivity of $H[U]$ (for $|U| \geq 2$), so we must have $Q_u \cup Q_v \subseteq U$. □

[1]Marco Bressan, Stefano Leucci, and Alessandro Panconesi. 2019. Motivo: Fast Motif Counting via Succinct Color Coding and Adaptive Sampling. *Proc. VLDB Endow. 12, 11 (July 2019), 1651–1663.* https://doi.org/10.14778/3342263.3342640

[2]Jianer Chen, Xiuzhen Huang, Iyad A. Kanj, and Ge Xia. 2006. Strong computational lower bounds via parameterized complexity. *J. Comput. System Sci. 72, 8 (2006), 1346 – 1367.* https://doi.org/10.1016/j.jcss.2006.04.007

Intuitively, the observation above states that U cannot "partially pick" a Q_v: as soon as U contains one vertex of Q_v, then it must contain the whole block. Similarly, U cannot contain a vertex c_e without also containing the whole blocks Q_u and Q_v.

Claim 3. *The construction above is an* FPT *reduction from* κ*-Clique to* κ*-SH.*

Proof. We prove that (G, k) is a YES-instance of κ-Clique if and only if (H, k') is a YES-instance of κ-SH. Recall that a YES-instance of κ-SH means that there exists $U \subseteq V(H)$ such that $|U| = k'$ and the section hypergraph $H[U]$ is connected.

(⇒) From κ-Clique to κ-SH. Assume that G contains a k-clique $K \subseteq V(G)$; define U as

$$U = \bigcup_{e=\{u,v\}\in E(G[K])} E_{uv} = \bigcup_{e=\{u,v\}\in E(G[K])} (Q_u \cup Q_v \cup R_e)\,, \tag{22}$$

where $G[K]$ is the subgraph induced by K. By construction, each vertex $v \in K$ contributes $\binom{k}{2}$ vertices through Q_v and each edge $e = \{u, v\} \in E(G[K])$ contributes exactly one vertex through R_e. Clearly, since K has $\binom{k}{2}$ edges, $|U| = k\binom{k}{2} + \binom{k}{2} = (k+1)\binom{k}{2} = k'$.

We now show that $H[U]$ is connected. For every edge $e = \{u, v\} \in E(G[K])$ we have $E_{uv} \subseteq U$, hence E_{uv} is an edge of $H[U]$. In particular, for every pair of distinct vertices $u, v \in K$ there is a hyperedge E_{uv} contained in $H[U]$ that includes all vertices of Q_u, all vertices of Q_v, and the vertex c_e corresponding to $e = \{u, v\}$; thus, $H[U]$ is connected.

(⇐) From κ-SH to κ-Clique. Assume that (H, k') is a YES-instance of κ-SH. Thus, there exists $U \subseteq V(H)$ such that $|U| = k'$ and $H[U]$ is connected. Intuitively, in what follows we decompose U into whole vertex blocks Q_v and edge vertices c_e, and then use counting and connectivity arguments to show that U must correspond to exactly k such blocks that are pairwise adjacent in G, that is, to a k-clique.

First, define T and S as follows

$$T = \{v \in V(G) : Q_v \subseteq U\} \text{ and } S = \{e \in E(G) : c_e \in U\}. \tag{23}$$

Each vertex $v \in T$ contributes all $\binom{k}{2}$ vertices of Q_v to U, and each edge $e \in S$ contributes the vertex c_e. By construction of H we have

$$V(H) = \bigcup_{v\in V(G)} Q_v \;\cup\; \bigcup_{e\in E(G)} R_e, \tag{24}$$

so every vertex of U lies either in some Q_v or in some $R_e = \{c_e\}$. Moreover, by Observation 1, if $v \notin T$ then $Q_v \cap U = \emptyset$, and if $c_e \in U$ then $e \in S$. Hence

$$U = \bigcup_{v\in T} Q_v \;\cup\; \{c_e : e \in S\}, \tag{25}$$

and we can express the size of U as

$$|U| = |T|\binom{k}{2} + |S|. \tag{26}$$

Call $t = |T|$ and $s = |S|$. Using $|U| = k' = (k+1)\binom{k}{2}$, we obtain

$$(k+1)\binom{k}{2} = t\binom{k}{2} + s \qquad \Longrightarrow \qquad s = (k+1-t)\binom{k}{2}. \tag{27}$$

Next, note that Observation 1 also implies that every edge $e = \{u, v\} \in S$ has both endpoints in T: indeed, since $c_e \in U$, the second item of Observation 1 gives $Q_u \cup Q_v \subseteq U$, so $u, v \in T$ by definition of T. Hence $S \subseteq \binom{T}{2}$. Let $G[T]$ be the graph with vertex set T and edge set S. By construction of H, the connected components of $H[U]$ correspond exactly to the connected components of $G[T]$: contracting each block Q_v to a single vertex v and deleting the vertices $\{c_e\}$ yields $G[T]$, and conversely every edge of $G[T]$ corresponds to a hyperedge E_{uv} in $H[U]$. Since $H[U]$ is connected, the graph $G[T]$ is connected. Therefore,

$$s \geq t - 1, \tag{28}$$

because any connected graph on t vertices has at least $t - 1$ edges. On the other hand, we clearly have

$$s \leq \binom{t}{2}. \tag{29}$$

We now derive constraints on t using Eqs. (27) to (29). In particular, we show that $t = k$, meaning that $s = \binom{k}{2}$ and thus G contains a k clique. We may assume $k \geq 3$, since for $k \leq 2$ both k-Clique and k-SH are trivial.

Case 1: $t \geq k + 1$. From Eq. (27) we obtain $s = (k + 1 - t)\binom{k}{2} \leq 0$. However, $H[U]$ is connected and $|U| = k' > 1$, so $H[U]$ must contain at least one hyperedge, which implies $s \geq 1$. This is a contradiction. Thus $t \leq k$.

Case 2: $t \leq k - 1$. From Eq. (27) we have $k + 1 - t \geq 2$, and hence

$$s = (k + 1 - t)\binom{k}{2} \;\geq\; 2\binom{k}{2} \;=\; k(k - 1). \tag{30}$$

On the other hand, since $s \leq \binom{t}{2}$ and $t \leq k - 1$, we have

$$k(k - 1) \leq s \leq \binom{k - 1}{2} \;=\; \frac{(k - 1)(k - 2)}{2}. \tag{31}$$

But clearly, for $k \geq 3$ we obtain

$$k(k - 1) > \frac{(k - 1)(k - 2)}{2}, \tag{32}$$

which leads to a contradiction. Therefore $t \not\leq k - 1$. The only remaining possibility is $t = k$. Finally, substituting into (27) we get

$$s = (k + 1 - k)\binom{k}{2} = \binom{k}{2}. \tag{33}$$

Since $s = |S|$ and $S \subseteq \binom{T}{2}$ with $|T| = k$, this means that S contains all possible $\binom{k}{2}$ edges on the vertex set T. Therefore $G[T]$ is a clique on k vertices, and hence G contains a k-clique. This is sufficient to prove the claim. □

Claim 4. *The construction above ensures that* $k' = (k + 1)\binom{k}{2} \leq \frac{k^3}{2}$ *and that* H *is* $(k^2, 0)$*-nice.*

Proof. First point follows simply by definition of the reduction; indeed we have $k' = (k + 1)\binom{k}{2} \leq \frac{k^2 \cdot k}{2} = \frac{k^3}{2}$ for $k \geq 1$.

We now show that H is $(k^2, 0)$-nice. By construction, every hyperedge of H is of the form

$$E_{uv} = Q_u \cup Q_v \cup R_e, \tag{34}$$

where $e = \{u, v\} \in E(G)$. Thus

$$|E_{uv}| = |Q_u| + |Q_v| + |R_e| = \binom{k}{2} + \binom{k}{2} + 1 = k(k - 1) + 1, \tag{35}$$

and hence

$$k(k-1)+1 = k^2 - k + 1 \leq k^2 \tag{36}$$

for every $k \geq 1$. Therefore the rank of H (the maximum size of a hyperedge) is at most k^2.

Consider the α-split of H with $\alpha := k^2$. Since no hyperedge has size greater than α, the upper part $H_{>\alpha}$ contains no hyperedges, and in particular every vertex has degree 0 in $H_{>\alpha}$. By the definition of (α, β)-niceness, this means that H is $(k^2, 0)$-nice. □

2 Additional Experiments

This section presents additional experimental results that complement the evaluation in Section 7 We extend the analysis along four main directions.

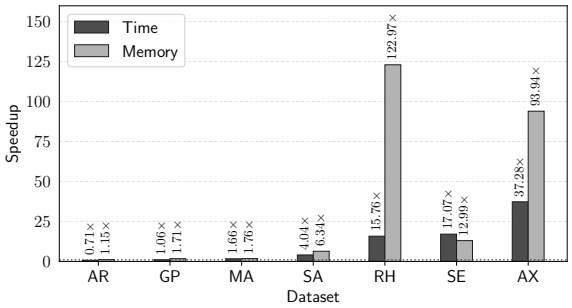
First, we provide further measurements of the build-up phase, including results for smaller hypergraphlet sizes ($k \in \{3, 4\}$), absolute running times, and a detailed breakdown of the build-up cost across datasets, which complement the speedup analysis reported in the main paper.

Second, we report a more detailed characterization of hypergraphlet frequency distributions, including the most frequent 4- and 5-hypergraphlets across all datasets.

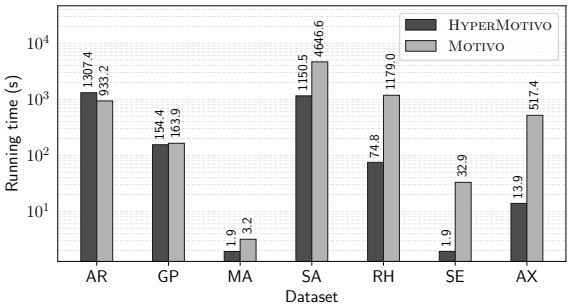
Third, we include an extended scalability study that evaluates the parallel performance of HyperMotivo across multiple datasets and values of k, covering a wider range of thread configurations and hypergraphlet sizes than those shown in the main paper.

Finally, we provide additional accuracy results, such as per-dataset error histograms and metrics for different sampling budgets and values of k, which refine and corroborate the empirical findings reported in Section 7.

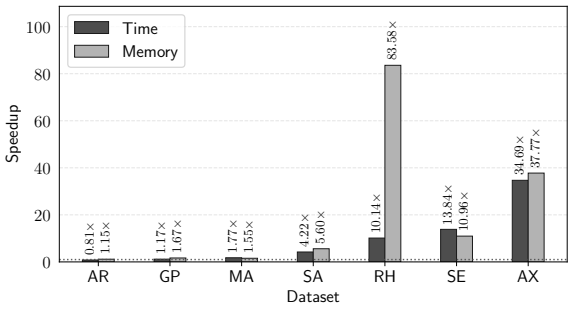
Overall, these experiments offer a more complete picture of the performance, (parallel) scalability, and accuracy of hypergraphlet counting with HyperMotivo.



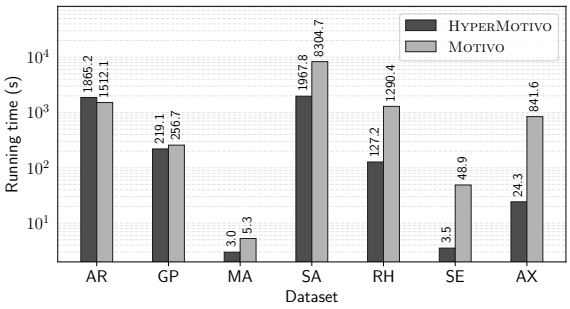
Time vs. memory ($k = 3$)



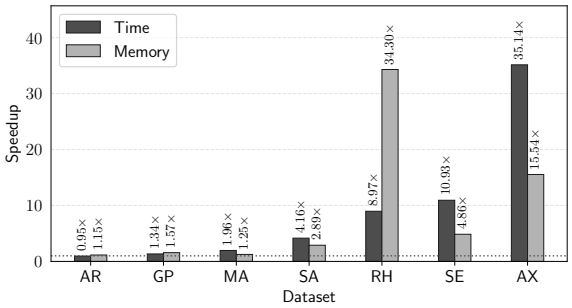
Absolute timings ($k = 3$)



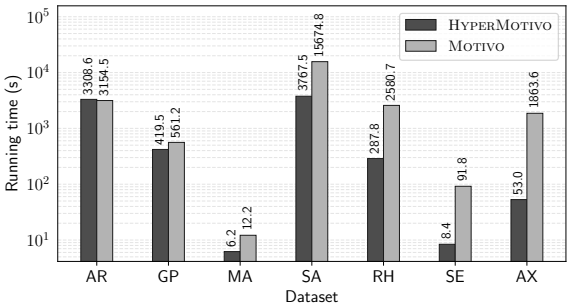
Time vs. memory ($k = 4$)



Absolute timings ($k = 4$)



Time vs. memory ($k = 5$)



Absolute timings ($k = 5$)

Figure 2: Memory–time tradeoff and absolute build-up timings for $k \in \{3, 4, 5\}$

2.1 Speedup and absolute build-up timings

Figure 2 shows, on the left, the speedup computed across all datasets. In addition to the case $k = 5$, which is already reported in the paper (see Section 7.2.2), we also include results for $k \in \{3, 4\}$. As observed in the paper, the speedup is consistent across different values of k. Higher values of k are computationally too demanding for the largest graphs. The right side of Figure 2 instead reports the absolute running times. Figure 3 reports absolute timing breakdown for each dataset and $k \in \{3, 4, 5\}$. For every dataset, the left group of bars reports the preprocessing and build-up times of Motivo, whereas the right group reports the preprocessing, build-up, and inclusion–exclusion times of HyperMotivo. For Motivo, preprocessing includes projecting the input hypergraph H to its Gaifman graph $\mathsf{Gaif}(H)$. For HyperMotivo, preprocessing includes selecting α, splitting H into $H_{\leq\alpha}$ and $H_{>\alpha}$, and projecting $\mathsf{Gaif}(H_{\leq\alpha})$.

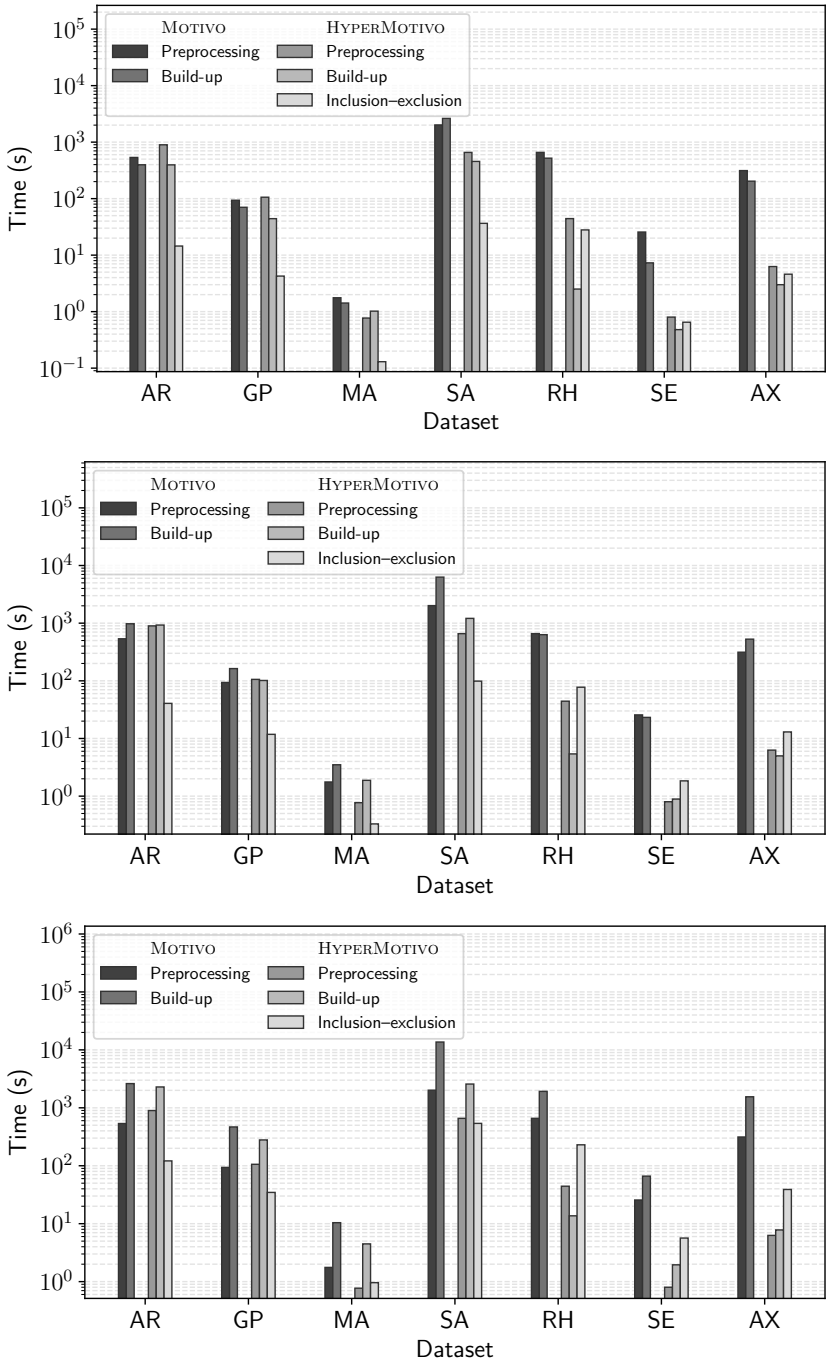
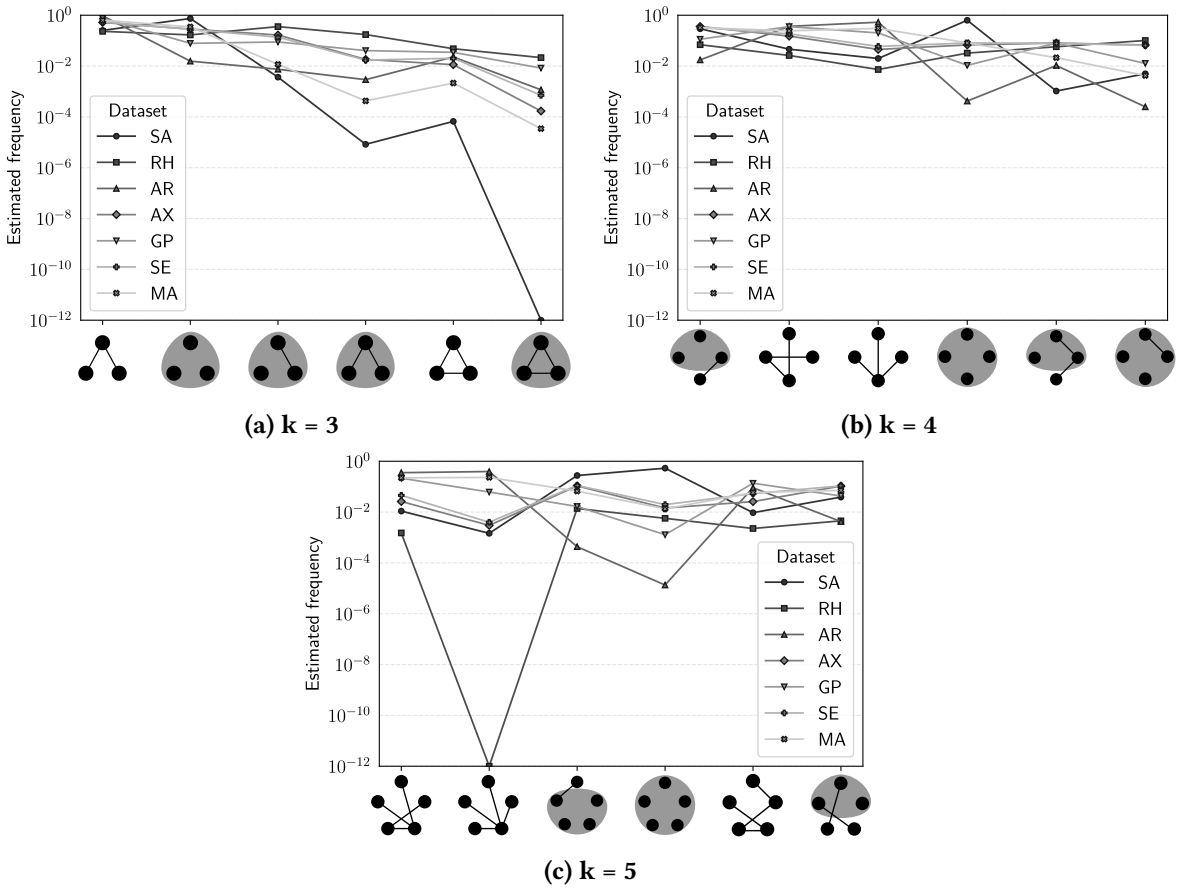


Figure 3: Timings breakdown for $k \in \{3, 4, 5\}$.



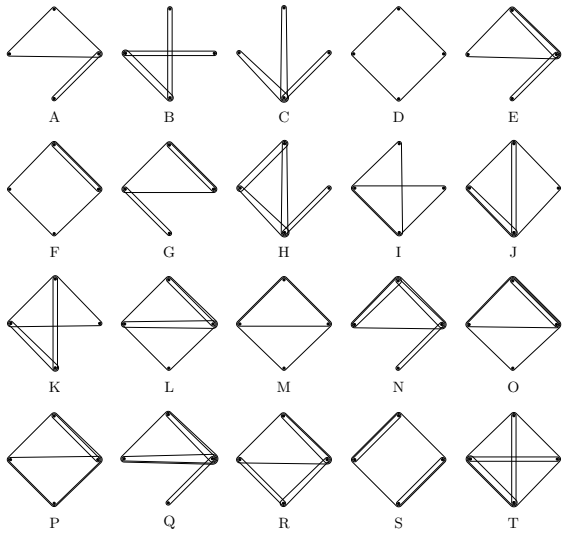
(a) k = 3 (b) k = 4

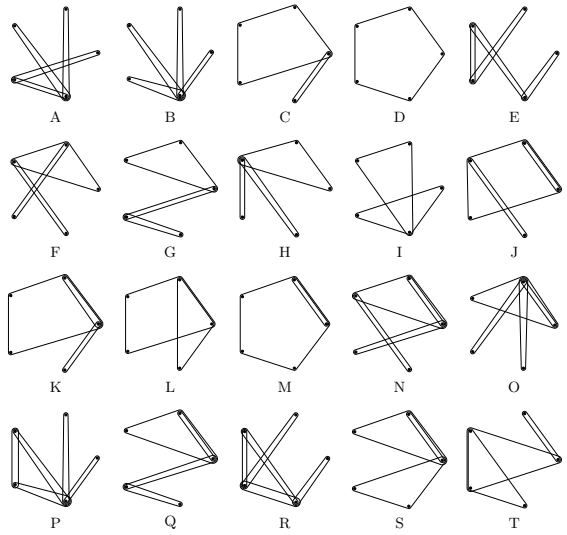
(c) k = 5

Figure 4: Frequency distributions of k-hypergraphlets for $k \in \{3, 4, 5\}$.

2.2 Hypergraphlet Frequencies

Figure 4 shows frequency distributions of $3, 4$ and 5-hypergraphlets with highest aggregate frequency across all hypergraphs. Figure 5 reports the frequencies of the 20 most frequent k-hypergraphlets for $k \in \{4, 5\}$ across all datasets.





	AR	AX	GP	SE	MA	RH	SA
A	0.0175	0.3605	0.1136	0.3281	0.3258	0.0692	0.2936
B	0.3649	0.1506	0.3564	0.1851	0.2424	0.0261	0.0461
C	0.5316	0.0450	0.2009	0.0592	0.3043	0.0073	0.0199
D	0.0004	0.0678	0.0107	0.0808	0.0814	0.0323	0.6324
E	0.0105	0.0806	0.0804	0.0790	0.0214	0.0575	0.0010
F	0.0002	0.0672	0.0127	0.0649	0.0042	0.1013	0.0050
G	0.0025	0.0515	0.0360	0.0461	0.0037	0.0287	0.0008
H	0.0543	0.0054	0.0487	0.0148	0.0082	0.0100	0.0001
I	0.0001	0.0515	0.0036	0.0280	0.0023	0.0300	0.0009
J	0.0001	0.0127	0.0060	0.0131	0.0002	0.0817	0.0000
K	0.0008	0.0230	0.0094	0.0288	0.0023	0.0298	0.0002
L	0.0001	0.0223	0.0052	0.0131	0.0002	0.0417	0.0000
M	0.0000	0.0173	0.0011	0.0061	0.0000	0.0442	0.0000
N	0.0029	0.0053	0.0297	0.0086	0.0006	0.0192	0.0000
O	0.0000	0.0049	0.0018	0.0029	0.0000	0.0426	0.0000
P	0.0000	0.0045	0.0010	0.0019	0.0000	0.0428	0.0000
Q	0.0029	0.0027	0.0204	0.0031	0.0010	0.0087	0.0000
R	0.0004	0.0025	0.0061	0.0064	0.0002	0.0202	0.0000
S	0.0000	0.0051	0.0015	0.0041	0.0000	0.0211	0.0000
T	0.0000	0.0008	0.0016	0.0012	0.0000	0.0266	0.0000

	AR	AX	GP	SE	MA	RH	SA
A	0.3549	0.0261	0.2148	0.0455	0.2222	0.0015	0.0109
B	0.3941	0.0030	0.0615	0.0040	0.2310	0.0000	0.0015
C	0.0004	0.1070	0.0166	0.1118	0.0667	0.0139	0.2747
D	0.0000	0.0144	0.0013	0.0197	0.0134	0.0057	0.5330
E	0.0908	0.0259	0.1360	0.0533	0.0584	0.0023	0.0095
F	0.0042	0.1055	0.0430	0.1071	0.0715	0.0045	0.0389
G	0.0052	0.0884	0.0425	0.0888	0.0872	0.0048	0.0430
H	0.0080	0.0447	0.0337	0.0465	0.1218	0.0025	0.0206
I	0.0001	0.0640	0.0029	0.0366	0.0303	0.0028	0.0541
J	0.0001	0.0418	0.0083	0.0402	0.0016	0.0159	0.0010
K	0.0003	0.0315	0.0093	0.0328	0.0043	0.0158	0.0016
L	0.0000	0.0419	0.0010	0.0210	0.0013	0.0131	0.0021
M	0.0000	0.0207	0.0019	0.0241	0.0010	0.0244	0.0069
N	0.0022	0.0185	0.0274	0.0205	0.0042	0.0033	0.0000
O	0.0074	0.0060	0.0276	0.0078	0.0239	0.0013	0.0002
P	0.0452	0.0004	0.0170	0.0012	0.0097	0.0000	0.0000
Q	0.0026	0.0110	0.0274	0.0198	0.0055	0.0045	0.0002
R	0.0305	0.0004	0.0181	0.0028	0.0062	0.0000	0.0000
S	0.0001	0.0266	0.0041	0.0172	0.0028	0.0032	0.0004
T	0.0000	0.0294	0.0026	0.0145	0.0011	0.0041	0.0004

Figure 5: First 20 4- and 5-hypergraphlets with highest aggregate frequency across all datasets. Left: K_4. Right: K_5.

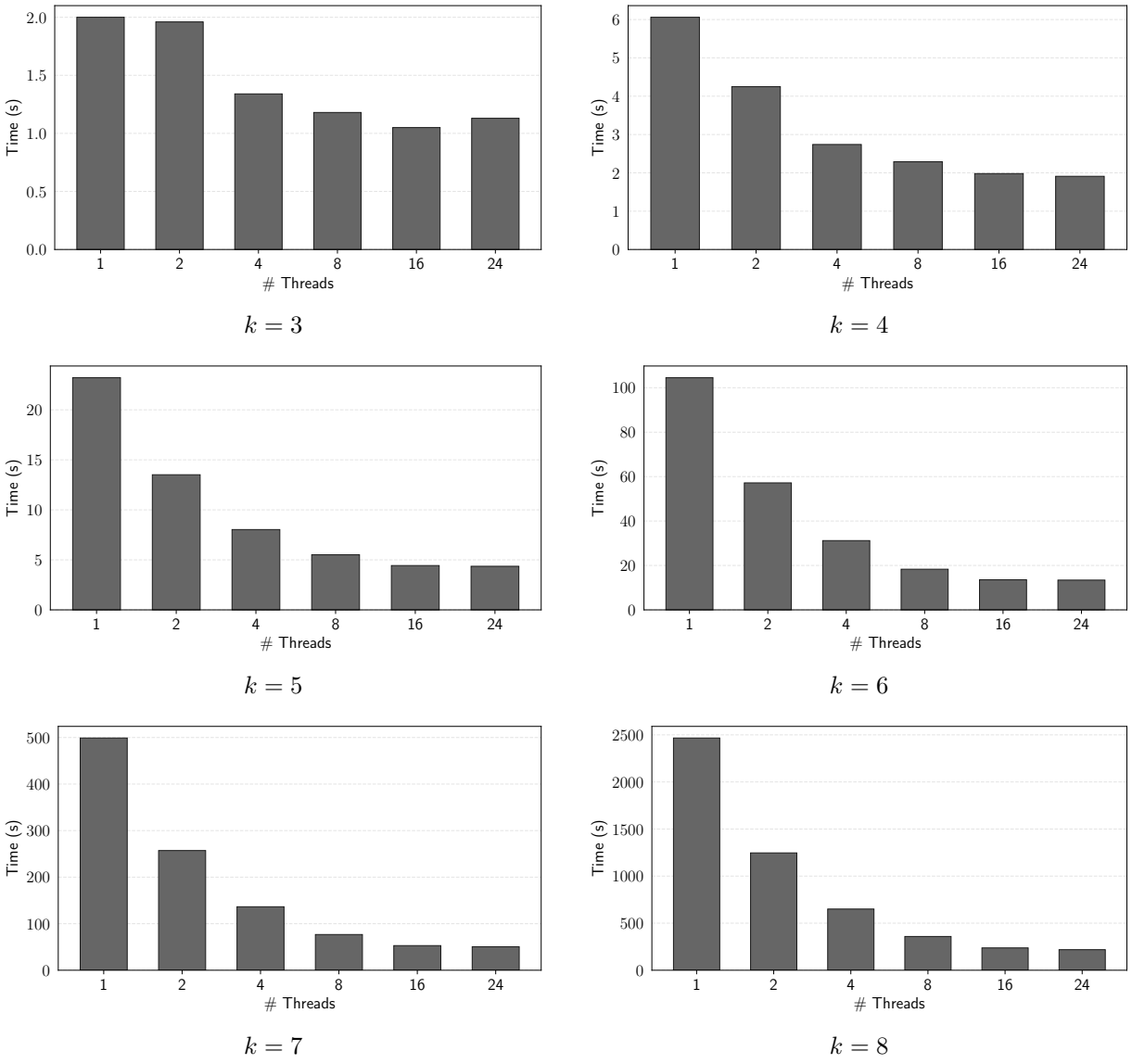


Figure 6: Running time of HyperMotivo's build-up phase on the MA dataset for $k \in \{3, \ldots, 8\}$.

2.3 Parallel scalability

Here we present parallel scalability experiments for the datasets MA (Figure 6), SE (Figure 7), and AX (Figure 8), for $k \in \{3, \ldots, 8\}$. We observe that scalability is consistent across all values of k and across all datasets. In some cases, when using 24 threads, a slight performance degradation is observed compared to 16 threads, likely due to communication overhead.

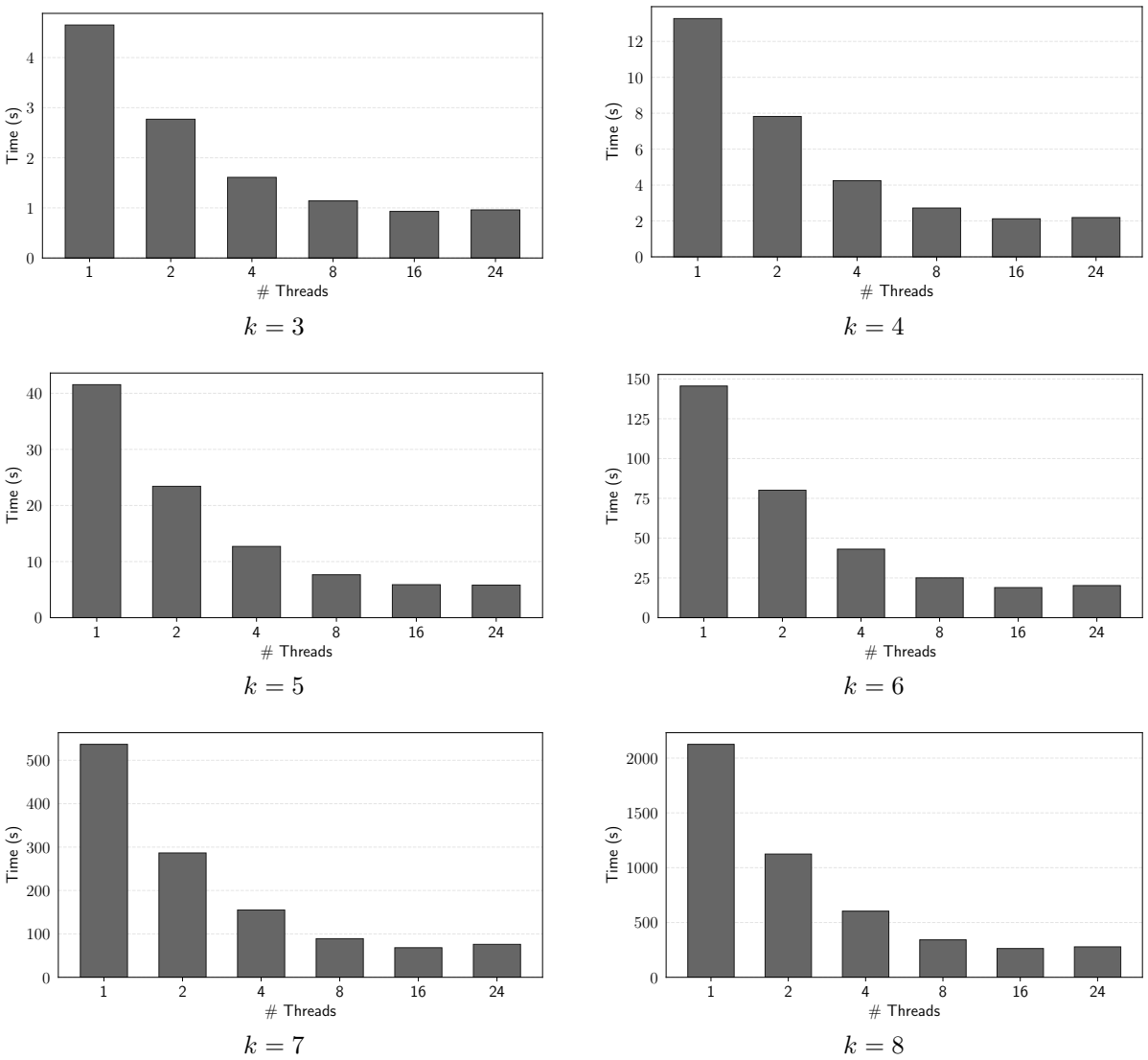


Figure 7: Running time of HyperMotivo's build-up phase on the SE dataset for $k \in \{3, \ldots, 8\}$.

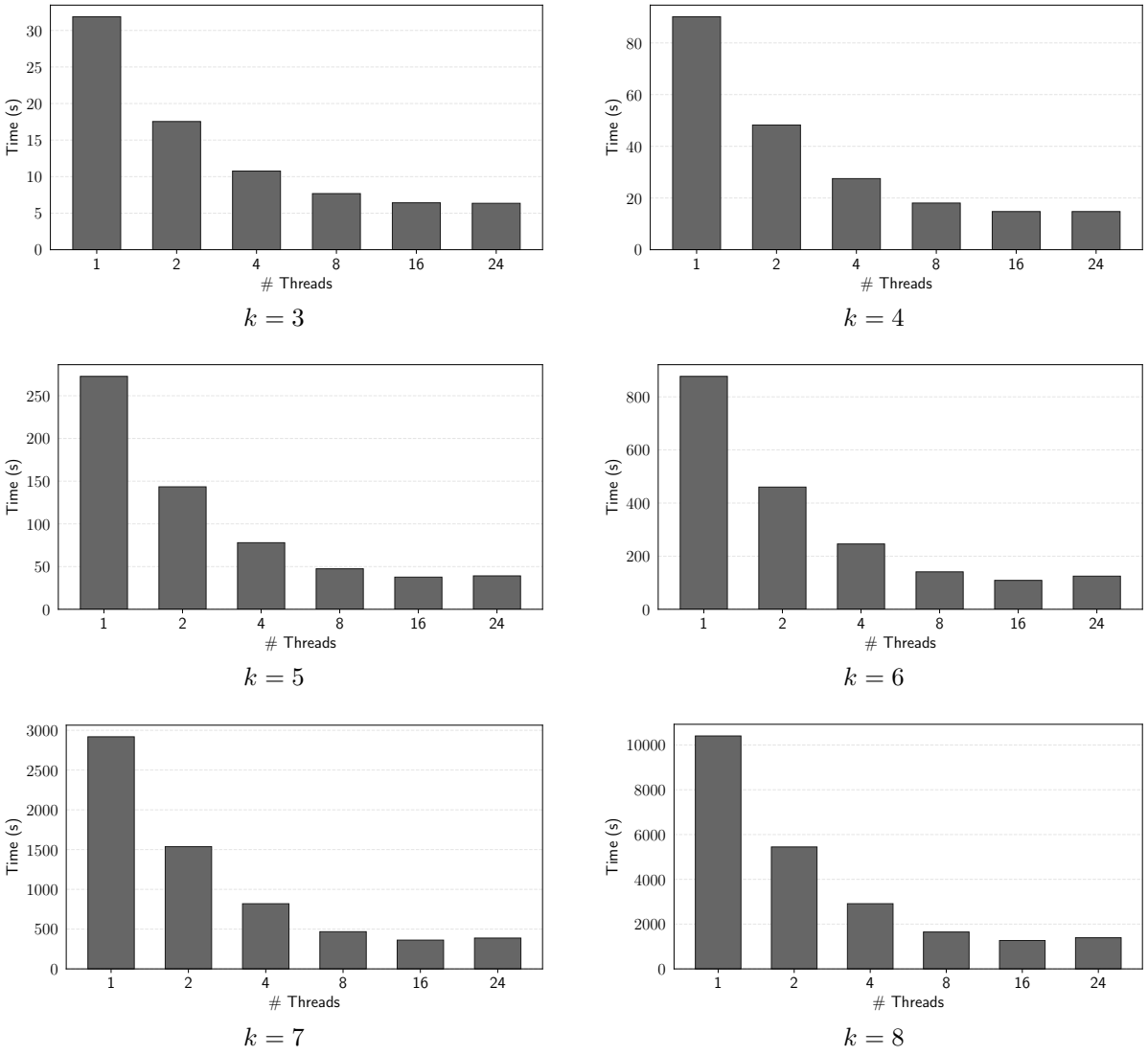
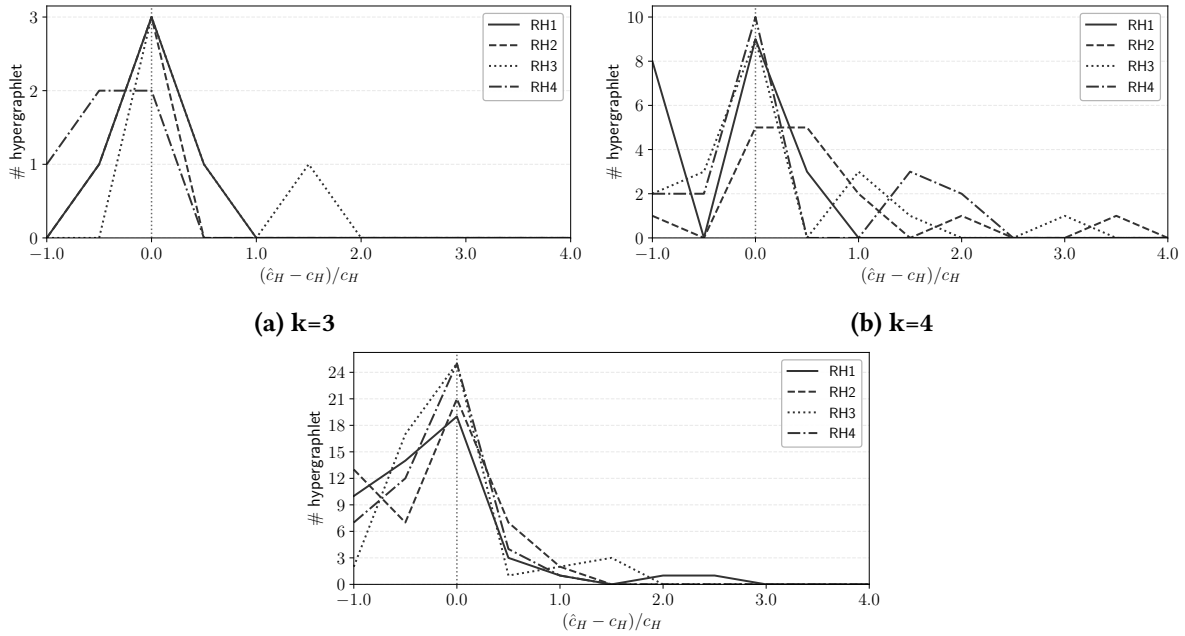


Figure 8: Running time of HyperMotivo's build-up phase on the AX dataset for $k \in \{3, \ldots, 8\}$.



(a) k=3 **(b) k=4**

(c) k=5

Figure 9: Count error err_H distributions for $k \in \{3, 4, 5\}$ and $S = 100000$ samples on four synthetic datasets.

2.4 Accuracy

Figure 9 shows distributions of per-hypergraphlet count errors $err_H = (\hat{c}_H - c_H)/c_H$, for $k = 3, 4, 5$ and $S = 100000$ samples. The setting is the same of the main paper: for every dataset, 50 runs are performed and the resulting estimates are averaged. For readability, the horizontal axis is truncated at 10: all hypergraphlets H with $err_H > 10$ are aggregated in the rightmost bin. Results for lower k shows a better accuracy by HyperMotivo; this has to be expected: the number of unique hypergraphlets (i.e, the number of isomorphism classes) grows drastically as k increases. This brings to a higher number of rare hypergraphlets, which are harder to find and estimate for the sampling algorithm, thus the higher count errors for higher k.